
What Does a Benign-Trained Anomaly Detector Actually Detect?

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Abstract

Anomaly detectors for insider threats are usually judged by how well their scores rank malicious activity above normal activity. We show that this number can be misleading, because it does not reveal what information the detector uses. We build a detector by adapting a large language model to normal employee activity only, so that unusual activity receives a high surprisal score, and we then open the model to determine what its score is based on: we isolate the internal changes that adaptation caused, approximate them with a sparse dictionary, test which dictionary directions causally influence the score, and measure what part of the input those directions respond to. On one standard benchmark the score turns out to rest almost entirely on who each user is (their recorded personality and organizational profile) rather than on what they did; on a second benchmark the score rests mostly on behavior. When malicious users are compared only against normal users that the model has never seen, apparent ranking quality falls from about 0.95 to near chance on the first benchmark. The finding survives an extensive robustness program: held-out users, repeated dictionary training, dictionaries trained without any malicious example, a smaller model retrained from scratch with fresh random seeds and a repaired data pipeline, and interventions whose strength is matched between treatment and control. The same analysis on a third dataset, collected from real people with real personality questionnaires, confirms the behavioral case, and classical statistical detectors trained on the same information do not take the profile shortcut. The conclusion is that high ranking performance alone does not show that a detector has learned to detect behavior, and this paper provides a way to check.

1 The problem

Organizations record what employees do on their computers: logons, file transfers, web activity, e-mail volume, and so on. An *insider threat* is an employee who misuses legitimate access, for example by stealing data. Because such events are rare, a common approach is *anomaly detection*: build a statistical model of normal activity, and flag days that the model finds unusual.

Formally, a detector is a function s that assigns a real-valued *anomaly score* $s(x)$ to each observation x , where in our setting x is a record of one user’s activity on one day. Higher scores are meant to indicate more suspicious activity. Detectors are usually evaluated by how well their scores *rank* malicious days above benign ones. The standard summary of ranking quality is the following probability, estimated over a labeled test set:

$$\text{AUC} = \Pr(s(X_{\text{mal}}) > s(X_{\text{ben}})), \quad (1)$$

where X_{mal} is a randomly chosen malicious day and X_{ben} a randomly chosen benign day. (This quantity is commonly called the ROC-AUC, for “area under the receiver operating characteristic curve”; the probability formulation is equivalent and is the only form we need.) An AUC of 1 means perfect ranking and 0.5 means the ranking is no better than chance.

The question that motivates our work is simple to state. A high AUC says that the score *separates* the two classes. It does not say *on the basis of what information* the separation is achieved. Two detectors with identical AUC could work very differently: one might respond to genuinely unusual behavior, while the other might respond to incidental properties of *who* the malicious users happen to be, such as their role, department, or personality profile. The second kind of detector would be nearly useless in practice, because in deployment the malicious users are not known in advance. This paper develops a method for determining which kind of detector one actually has, applies it to a modern detector built from a large language model, and then subjects the answer to a systematic robustness program.

2 The data

We use two versions (called r4.2 and r6.2) of the CERT insider-threat corpus, a widely used synthetic dataset that simulates a corporation’s activity logs, produced by Carnegie Mellon University. Each version contains several thousand simulated employees observed over about eighteen months, with a small set of employees scripted to carry out malicious scenarios on particular days. Version r4.2 has sixty malicious users among about one thousand; version r6.2 has four among about four thousand. This difference in composition turns out to matter, and we return to it repeatedly. A third dataset, TWOS, collected from real people, is described in Section 7.

Each user-day is converted into a short structured text with three kinds of lines:

- a **DAY** line holding static organizational fields for that user (project, role, business unit, department, team, and an information-technology administrator flag);
- a **PSY** line holding the user’s five simulated personality scores. These are the “Big Five” personality traits used in psychology: openness (O), conscientiousness (C), extraversion (E), agreeableness (A), and neuroticism (N). In CERT these numbers are generated by the simulator and are constant for each user;
- one **SES** line per work session, holding numeric behavioral counts for that session: duration, number of logons, file and removable-drive activity, e-mail volume, web-browsing category counts, and similar quantities.

A user-day text is therefore a small table serialized as text. The first two lines describe *who the user is* and are essentially constant across a user’s days; the remaining lines describe *what the user did* that day. We call the first two line classes *profile* content and the third *behavioral* content. This partition of the input is the reference frame for the entire paper.

Users are split as follows. Users with any malicious day are reserved entirely for evaluation. The remaining benign users are divided into a training set (roughly ninety percent) and a validation set (roughly ten percent) whose days never enter training. The validation users will play an important role: they are the only benign users about whom the trained model knows nothing.

3 The detector

3.1 Language models as probability distributions

A *language model* is a parametric probability distribution over sequences of symbols called tokens. Writing a sequence as $x = (x_1, \dots, x_n)$, the model has the product form

$$p_\theta(x) = \prod_{t=1}^n p_\theta(x_t \mid x_1, \dots, x_{t-1}), \quad (2)$$

where each conditional distribution is computed by a neural network with parameters θ . Modern large language models are such networks with billions of parameters, trained on enormous general text corpora.

We use one called Qwen3-8B (eight billion parameters). The network processes a sequence through L successive layers; at layer ℓ and position t it maintains an internal state vector $h_t^{(\ell)} \in \mathbb{R}^d$ with $d = 4096$. These internal vectors matter later: they are the objects we analyze.

3.2 Adapting the model to normal behavior only

We specialize the general-purpose model to our corpus by *fine-tuning*: continuing its training on the serialized user-day texts, using only days from benign training users. The objective is maximum likelihood, that is, minimizing the negative log-likelihood of the benign corpus $\mathcal{D}$:

$$\mathcal{L}(\theta) = - \sum_{x \in \mathcal{D}} \sum_t \log p_\theta(x_t | x_{<t}). \quad (3)$$

No malicious example is ever used in training. This is the classical one-class setup: model the normal population, then measure deviation from it.

For efficiency the fine-tuning does not modify all eight billion parameters. Each weight matrix W of the network is kept fixed and receives a trainable low-rank correction,

$$W' = W + \frac{\alpha}{r} BA, \quad B \in \mathbb{R}^{d_{\text{out}} \times r}, \quad A \in \mathbb{R}^{r \times d_{\text{in}}}, \quad (4)$$

with rank $r = 16$, so only the small matrices A and B are learned. This technique is called LoRA (low-rank adaptation); we use a memory-saving variant called QLoRA in which the fixed weights are stored at reduced numerical precision. The pair (fixed base model, learned low-rank correction) is called the *adapted model*. The important point is that the entire effect of training is carried by a low-dimensional, removable perturbation, which makes before-versus-after comparisons clean.

3.3 The anomaly score and two ways to evaluate it

The score of a day x is its average surprisal under the adapted model:

$$s(x) = -\frac{1}{n} \sum_{t=1}^n \log p_{\theta_{\text{ada}}}(x_t | x_{<t}). \quad (5)$$

Days that the model of normal behavior finds improbable receive high scores.

There are two reasonable ways to estimate the AUC of equation (1), and the difference between them will turn out to be one of the paper’s findings.

1. *The standard protocol.* Malicious days are compared against benign days drawn from the general population. Because the model was trained on about ninety percent of benign users, most of those benign comparison days belong to users the model saw during training. Under this protocol the adapted model looks excellent: AUC about 0.953 on r6.2 and 0.964 on r4.2, higher than four standard anomaly-detection baselines evaluated on the same benchmarks (an isolation forest, a deep one-class method called Deep SVDD, and two recurrent autoencoders), whose AUC values range from about 0.63 to 0.77.
2. *The user-disjoint protocol.* Malicious days are compared only against benign days of the validation users, whom the model never saw. Malicious users are unseen by construction, so under this protocol both sides of the comparison are equally unfamiliar to the model.

The gap between these two numbers measures how much of the apparent performance is explained by mere familiarity with specific users rather than by properties of the activity itself. We report the gap in Section 5.2.

4 Opening the model

We now describe the four analysis steps, in the order they are applied. Together they take us from the raw network to a statement about what the score is based on.

4.1 Step 1: isolate what fine-tuning changed

Because the base model is kept frozen, we can run the same day x through both the base and adapted models and subtract their internal states. Define, at a chosen layer ℓ and each token position t ,

$$\delta_t = h_t^{(\ell),\text{ada}} - h_t^{(\ell),\text{base}} \in \mathbb{R}^d. \quad (6)$$

The vector δ_t isolates the representational change induced by the adaptation at the audited layer and position. It is the natural object to study if one wants to know what the adaptation, as opposed to the general-purpose model, contributes at that point in the computation.

4.2 Step 2: approximate the changes with a sparse dictionary

The vectors δ_t live in $\mathbb{R}^{4096}$ and are not individually interpretable. We therefore approximate them by sparse nonnegative combinations of learned dictionary elements. After standardizing each coordinate (subtracting its mean μ and dividing by its standard deviation σ), we seek a dictionary matrix $D \in \mathbb{R}^{d \times m}$ (with $m = 4d$ columns, called *features*) and, for each standardized vector $\tilde{\delta}_t$, a coefficient vector $z_t \in \mathbb{R}_{\geq 0}^m$ with at most $k = 8$ nonzero entries, such that

$$\tilde{\delta}_t \approx Dz_t, \quad (7)$$

with D and the coefficients chosen to minimize the mean squared reconstruction error over a large sample of token positions. This is a sparse coding problem; the particular parametrization we use (a one-layer encoder network followed by a keep-the-largest- k rule and a linear decoder) is called a *sparse autoencoder* in the machine-learning literature. The outcome is that each change vector is expressed as a short combination of reusable directions, and each direction can be studied on its own. One technical point deserves notice: because the standardization constants differ between datasets, any geometric comparison of features across datasets must first map the dictionary columns back to the original coordinates, that is, compare the directions $\sigma \odot D_{:,f}$ rather than the raw columns. All cross-dataset comparisons in this paper do so.

4.3 Step 3: select the score-relevant features and test them causally

Some dictionary features activate similarly on malicious and benign days; those cannot carry the anomaly signal. We rank features by the difference in mean activation,

$$g_f = \mathbb{E}[z_{t,f} \mid \text{malicious day}] - \mathbb{E}[z_{t,f} \mid \text{benign day}], \quad (8)$$

and take the five features with the largest g_f as the candidate set S . As a comparison group we also select a matched *control* set C of five features that are similarly active overall but carry no class difference (small $|g_f|$). Selection by a statistic can be flattered by chance, so everything that follows treats S as a hypothesis to be tested rather than a conclusion.

The test is interventional. The idea is to edit the coefficients of the selected features inside the model, let the edited internal state flow through the remaining layers, and observe whether the anomaly score responds. We use two complementary edits.

Repair (a sufficiency-style test). For a malicious day, we move the selected coefficients toward the values they take on a comparable benign day (a “donor” from the same organizational context, never the same user), with an interpolation strength $\alpha \in \{0.25, 0.5, 0.75, 1\}$, and record the best achievable reduction of the score over the available donors and strengths. Write Δ_{ben}^S for this quantity, and Δ_{anom}^S for the analogous quantity when donors are other malicious days. If the selected features carry anomaly-specific repairable structure, moving them toward benign values should reduce the score more than moving them toward other anomalous values, and it should do so more for S than for the control set C . The reported estimand is the doubly differenced quantity

$$\tau = (\overline{\Delta_{\text{anom}}^S} - \overline{\Delta_{\text{ben}}^S}) - (\overline{\Delta_{\text{anom}}^C} - \overline{\Delta_{\text{ben}}^C}). \quad (9)$$

The double difference protects the comparison: any effect that comes merely from making a large edit, rather than from the specific content of the selected features, appears in both arms and cancels at the level of the contrast.

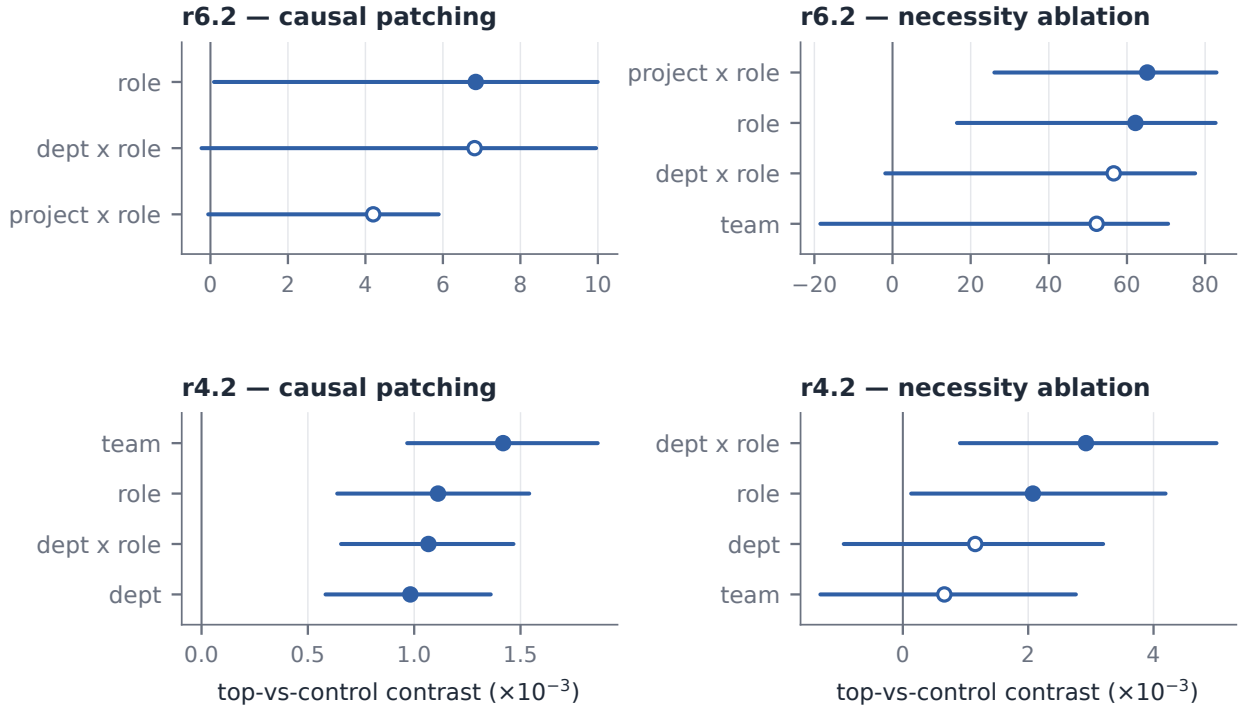


Figure 1: Effect of editing the selected features, compared with editing matched control features, on the anomaly score (one panel per benchmark and per type of edit; units are thousandths of the score). Each point is the average effect for one organizational matching context, and each horizontal bar is a 95 percent confidence interval computed with the malicious users as the resampling unit. Positive values mean that the selected features influence the score more strongly than the control features do. Open circles mark intervals that include zero.

Ablation (a necessity-style test). We scale the selected coefficients toward zero, on malicious days and on matched benign days, and ask whether removal disrupts the score more on the malicious side and more for S than for C . The corresponding double difference is denoted ν .

Two safeguards are essential and were arrived at the hard way. First, donors from the same user are excluded: allowing them lets the procedure “repair” a malicious day by borrowing the same person’s own benign signature, which inflates effects for reasons unrelated to the question. Second, uncertainty is computed by resampling *users*, not days: multiple days of the same user are statistically dependent, so all confidence intervals in this paper are user-level bootstrap intervals with the malicious user as the resampling unit (10,000 draws). On r6.2, which has only four malicious users, such intervals are reported as descriptive.

One further check matters because the selected features are, by construction, strongly activated: their edits are physically larger than edits of the control features. We therefore repeated the headline comparisons with *dose-matched* controls, chosen to match the selected features on activation frequency, activation magnitude, and dictionary-direction norm, which closes the edit-size gap to a factor of 1.3 to 1.8. The contrasts survive: on r4.2 the repair contrast remains clearly positive (+0.0017 with interval [0.0011, 0.0023]) and on r6.2 the ablation contrast remains clearly positive (+0.061 with interval [0.007, 0.084]); the remaining two contrasts stay positive but with intervals containing zero. Figure 1 summarizes the intervention measurements.

4.4 Step 4: read off what the features respond to

Finally we ask *where* the selected features activate within the day text. Every token position belongs to one line class: the organizational DAY line, the personality PSY line, or a behavioral SES line. For a feature f , define its *attribution mass* on class c as the fraction of its total activation that occurs at positions of that

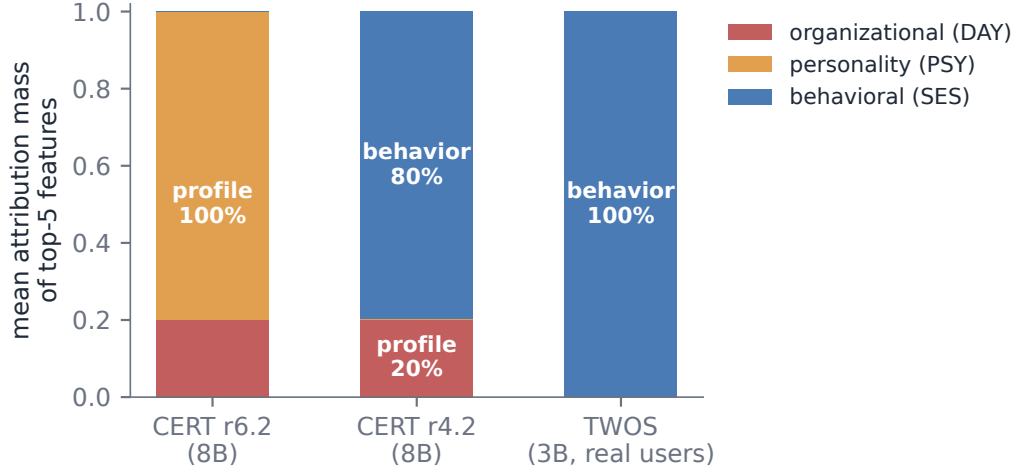


Figure 2: Where the selected features place their activation, averaged over each detector’s five selected features. Each bar decomposes the activation mass into the organizational lines, the personality lines, and the behavioral session lines of the input text. The r6.2 detector’s evidence is almost entirely a description of who the user is; the r4.2 and TWOS detectors’ evidence is almost entirely a description of what the user did.

class:

$$m_f(c) = \frac{\sum_{(x,t) : \text{class}(t)=c} z_{t,f}(x)}{\sum_{(x,t)} z_{t,f}(x)}. \quad (10)$$

If a feature has nearly all of its mass on PSY and DAY positions, it responds to who the user is; if its mass lies on SES positions, it responds to what the user did.

The logic of the whole method is a conjunction. The interventions of Step 3 establish that the selected features are causally connected to the score, but say nothing about their meaning. The attribution of Step 4 establishes what the features respond to, but says nothing about whether they matter. Only together do the two steps answer the question in the title, and the paper’s findings repeatedly illustrate that either step alone would have been misleading.

5 Findings on the two CERT benchmarks

5.1 Two detectors that look alike and are not

The two CERT versions produce nearly identical headline AUC values, yet the analysis reveals qualitatively different detectors.

On r6.2, the five selected features place between 99.8 and 100 percent of their attribution mass on the personality and organizational lines, with activation on the personality line enriched about thirteen times over its share of tokens. The intervention contrasts are positive (repair $\tau = 0.0068$ with descriptive interval $[0.0001, 0.0100]$; ablation $\nu = 0.065$ with interval $[0.026, 0.083]$). The detector’s causally relevant evidence is almost entirely a description of *who the user is*.

On r4.2, four of the five selected features concentrate on the behavioral session lines (the fifth responds to the organizational header). The contrasts are again positive (repair $\tau = 0.0014$ with interval $[0.0010, 0.0019]$; ablation $\nu = 0.0029$ with interval $[0.0009, 0.0050]$). There, the evidence is mostly *what the user did*. Figure 2 shows the contrast, together with the real-data result of Section 7.

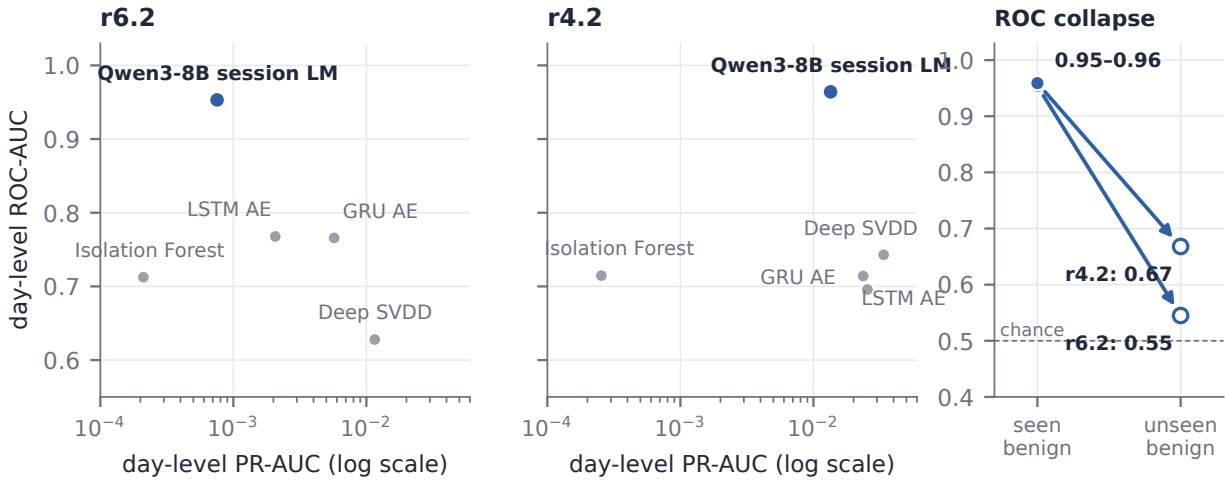


Figure 3: Score-level results. Left and center panels: ranking quality (vertical axis) and precision (horizontal axis, logarithmic scale) for the language-model detector (blue) and four standard anomaly-detection methods on the two benchmarks; the language model attains the highest ranking quality under the standard protocol. Right panel: the same language-model ranking quality when malicious users are compared only against benign users never seen during training. The apparent advantage largely disappears, and the dashed line marks chance level.

5.2 The consequence for the headline numbers

Why would reading a user’s static profile produce a high AUC at all? The adapted model was trained on roughly ninety percent of the benign users. For those users, the model has been repeatedly exposed to the constant profile lines, so their profile text is comparatively familiar and their days look unsurprising. The malicious users were never seen in training, so their profile lines alone make their days look surprising. The score is then partly measuring *familiarity* rather than behavior.

The user-disjoint protocol of Section 3.3 removes this asymmetry, and the effect is large. The AUC falls from 0.954 to 0.545 on r6.2, which is barely better than chance, and from 0.959 to 0.668 on r4.2. Under the fair comparison the language model no longer outperforms the classical baselines at all. Two further measurements complete the picture. First, ranking each malicious user’s own days (their malicious days against their own benign days, identity held fixed) still succeeds moderately on both benchmarks (average within-user AUC 0.67 on r4.2 and 0.73 on r6.2), so behavioral signal exists; it is simply drowned, in the cross-user comparison, by the familiarity effect. Second, deleting the personality lines from the inputs at scoring time *improves* the r4.2 detector’s performance on unseen users from 0.67 to about 0.80 (an increase of 0.127 with interval [0.061, 0.193]): the profile text was not merely unhelpful, it was actively obscuring the behavioral signal. On r6.2, deleting profile content changes nothing that is statistically distinguishable; with every user unseen, profile novelty separates nothing, and no clear behavioral signal is revealed underneath. Figure 3 shows the baseline comparison and the collapse.

6 Is the interpretation robust?

A single analysis pipeline contains many arbitrary choices, and each is an opportunity for an artifact. This section walks through the checks, one question at a time. Each check either survived or, where it did not, the failure is reported and incorporated into the claims.

Does the finding survive on users that played no role in selecting the features? On r4.2 we split the sixty malicious users into two halves, re-selected features using only the first thirty, and estimated effects only on the second thirty. Four of the five selected features reappear, and all effect estimates remain positive on the

held-out half, at roughly half the original magnitude, with one matching context individually excluding zero. The shrinkage is the expected signature of selection effects, and the claims are scoped accordingly. On r6.2, with only four malicious users, we used leave-one-user-out selection instead; there the effects concentrate on the single dominant user, who contributes 46 of the 70 malicious days, and the paper reports r6.2 causal claims at that resolution.

Is the dictionary itself an accident of its random initialization? Training the dictionary three times with different random seeds and comparing the resulting feature directions (in original coordinates, as noted in Step 2) shows that each benchmark’s selected feature family is stable: matched features across seeds align with cosine similarity 0.81 to 0.95, above a null baseline of 0.72 to 0.82 computed from the full dictionaries. Across benchmarks, by contrast, the selected features align at or below the null; notably, where the two dictionaries share the most overall geometry, the selected features align *worse* than typical features. The two detectors’ evidence is not the same structure under different labels.

Does the finding depend on malicious examples having shaped the dictionary? The dictionaries above were fitted on a pool that includes malicious days. We retrained both dictionaries from scratch on benign rows only, so that no malicious example influences dictionary formation, and repeated selection, attribution, and evaluation. The attribution dissociation reproduces exactly (r4.2 selected features at least 94 percent behavioral; r6.2 features at least 99 percent profile-bound in every leave-one-user-out fold). The intervention results are estimand-dependent under the new dictionaries: the ablation contrast on r4.2 replicates across all matching contexts, while the repair contrast does not. We therefore treat ablation dependence as the robust causal signature and report repair effects with that qualification.

Is the difference between the benchmarks simply that one has four malicious users and the other sixty? We subsampled r4.2’s malicious population down to four users (ten random repetitions per size) and re-ranked features each time. The profile share of the selected features stays low and nearly flat across population sizes. Raw positive count alone does not explain the dissociation; the composition of the population, not its size, is what matters.

Does the finding survive a fresh model, fresh randomness, and a repaired pipeline? A data-serialization defect (four file-transfer fields silently dropped) was discovered during the project. We repaired it and reran the entire pipeline at smaller scale: a different, smaller language model (Qwen2.5-3B) trained from scratch twice per benchmark with different random seeds, new dictionaries trained on benign rows only, and fresh selection and attribution. The dissociation reproduces under both seeds: r4.2 selected features are behavioral in every audited configuration, and r6.2 develops features that are 100 percent profile-bound at the deepest audited layer under both seeds. The ablation contrast again replicates on r4.2 across seeds (+0.0022 and +0.0021) while the repair contrast again does not, matching the pattern above.

Could the score’s profile sensitivity be an illusion of the analysis rather than a property of the score? Two input-level experiments require no interpretability machinery at all and agree with the internal analysis. Replacing a malicious day’s profile lines with a random benign user’s profile changes the r6.2 score substantially, while replacing its entire behavioral content changes it much less; on r4.2 the sensitivities are reversed. And the masking result of Section 5.2 shows the score-level consequence directly.

7 Real data with real personality scores

The CERT data are simulated, including the personality scores. The natural objection is that the entire phenomenon might be an artifact of simulation. We therefore repeated the pipeline on TWOS, a public dataset collected at the Singapore University of Technology and Design from twenty-four real users over one week, during which, by design of the underlying study, twelve users had their sessions temporarily taken over by other participants (a “masquerade” attack), and five users, after a staged dismissal, briefly acted as insiders on their own accounts (a “traitor” scenario). Each real user completed the standard fifty-item Big Five questionnaire, so the serialized inputs contain genuine psychometric information.

The details mirror the CERT setup at smaller scale. Mouse, keyboard, and logon telemetry are aggregated into ten-minute windows and serialized in the same three-line-class format, with the real Big Five scores forming the personality line (18,048 windows in total, of which 138 overlap a staged attack: 101 masquerade

and 37 traitor windows). The model is the smaller Qwen2.5-3B, adapted on benign windows only (about seven thousand windows from the twelve users who were never masqueraded; for the five later dismissed users, only days before the dismissal enter training), trained twice with different random seeds; dictionaries are trained on benign rows only; selection and attribution proceed exactly as before.

Because half of the users are attacked, profile information is a far weaker separator here than in CERT r6.2, and our account predicts that the detector should rely on behavior. That is what the analysis finds. The selected features place essentially all of their attribution mass on the behavioral lines, under both training seeds and at both audited layers (the smallest behavioral share of any selected feature is 0.983). Decisively, the evaluation compares each attacked user’s attack windows with that same user’s normal windows, so static identity and personality information is held fixed and cannot explain the observed separation; the average within-user AUC is 0.78 across the sixteen attacked users (the twelve masqueraded users plus the four dismissed users with logged activity in their staged window). The intervention experiments reproduce the pattern seen on CERT: the ablation contrast is positive (+0.0019) while the repair contrast is not, the fourth independent occurrence of that asymmetry. The behavioral pole of the dissociation is not an artifact of synthetic data or simulated psychometrics.

8 Which detectors take the shortcut?

One further question separates two very different readings of the r6.2 result. Is the profile information so predictive in that dataset that *any* statistical method would exploit it, or is the shortcut a property of this particular detector’s construction?

To answer it we trained three classical anomaly detectors on the same r6.2 days, represented as ordinary numeric feature vectors that *include* the profile variables (twelve profile columns among 242 behavioral ones): an isolation forest, a one-class support vector machine, and a principal-component reconstruction method, each trained only on benign training users and evaluated user-disjointly, exactly as the language model was. For attribution we use permutation importance, which is defined without reference to any particular model class: for each input column, randomly permute that column’s values across the evaluation days, recompute the AUC, and record the decrease; the importance of a group of columns is its share of the total decrease.

All three classical detectors essentially ignore the profile variables. The profile columns receive four to seven percent of the total importance, no profile column appears among any detector’s five most important columns, and deleting the profile columns entirely does not reduce any detector’s AUC (it slightly increases it, by 0.003 to 0.022). The same holds on r4.2. The shortcut is therefore not an inevitable consequence of the available profile information: we do not observe comparable profile reliance in the three classical baselines. An explanation consistent with this evidence lies in the training objective. For a likelihood-based sequence model, the per-user-constant profile tokens are the most predictable part of every training sequence, so learning them yields a large and cheap reduction of the training loss; in a plain feature-vector representation the same information is twelve columns among 242, with no comparable incentive attached. This also explains a number from Section 5.2: under the fair protocol the language model *underperforms* the classical baselines on r6.2, exactly as expected if its adaptation spent capacity on a shortcut that does not generalize while the baselines learned behavior.

9 Conclusions

The overall picture can be stated as a two-factor account. The composition of a dataset’s malicious population determines *when* an identity shortcut is available: profile information separates the classes when the malicious users are a small, distinctive minority (CERT r6.2) and separates nothing when they are not (TWOS). The training objective and input representation determine *which* detectors take an available shortcut: the likelihood-based sequence model took it, and three classical feature-vector detectors did not. Both factors were tested rather than assumed: the first by subsampling and by an out-of-setting prediction that the real-user dataset confirmed, the second by the cross-detector comparison.

Three practical conclusions carry beyond these datasets and this model. First, for anomaly detectors intended to detect behavior, ranking quality alone is not evidence of success; evaluation should compare malicious observations only against normal observations from equally unseen users, and, where possible, within users. Second, including static descriptors of people (role, department, personality measures) in a detector’s input is not harmless; here it created the shortcut in one detector and degraded performance even where the detector was behavioral. Third, the question “what does this detector actually detect?” is answerable: differences of internal states, a sparse dictionary approximation, causal testing of selected directions against matched controls, and attribution of those directions to input content jointly turn the question into a measurable quantity, and the robustness program of Section 6 is what entitles the answer to be believed.

The claims are scoped by design. The causal results concern the particular adaptation runs audited here; the r6.2 estimates rest on four malicious users dominated by one; the CERT data are synthetic; and the real-data study is small. What survives every variation we tested is the central dissociation and its explanation, and the method by which both were found.